



ELECTRICAL MEASUREMENTS

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ELECTRICAL MEASUREMENTS

Scope of presentation

Introduction

Types of attenuators

1. Fixed Type Attenuator
2. Step Type Attenuator
3. Continuously Variable Type Attenuator
4. Programmable Type Attenuator
5. Bias Type Attenuator
6. DC Blocking Type Attenuator
7. Optical Attenuators

Attenuators Circuit

Attenuator characteristics

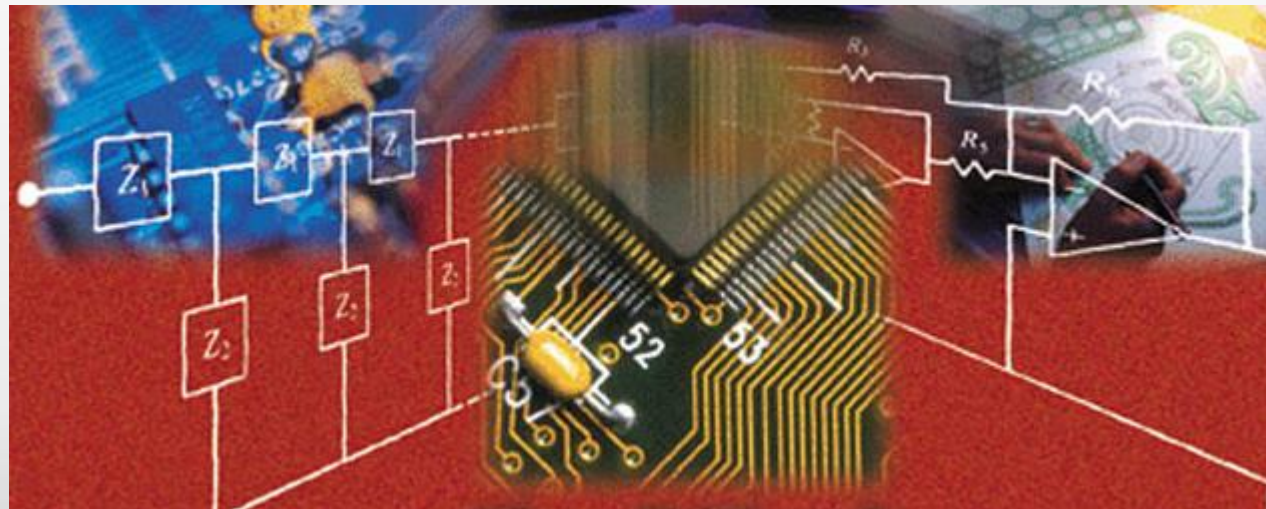
Application of the attenuators

Switched Attenuator

Thanks & Questions



Menoufia University FACULTY OF Electronic Eng



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- ✓ *An attenuator is an electronic device that reduces the power of signal without appreciably distorting its waveform.*
- ✓ *As we know, an attenuator is effectively the opposite of amplifier.*
- ✓ *Attenuators are usually passive devices made from simple voltage divider networks.*

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- Attenuators are available as both fixed attenuators and adjustable attenuators.
- Fixed attenuators networks are known as "attenuator Pads".
- These are available for specific values from 0dB to 100dB.
- Adjustable attenuators is a circuit that decreases the A variable attenuator is a circuit that decreases
- the strength of the input signal either continuously or step by step without appreciable signal distortion

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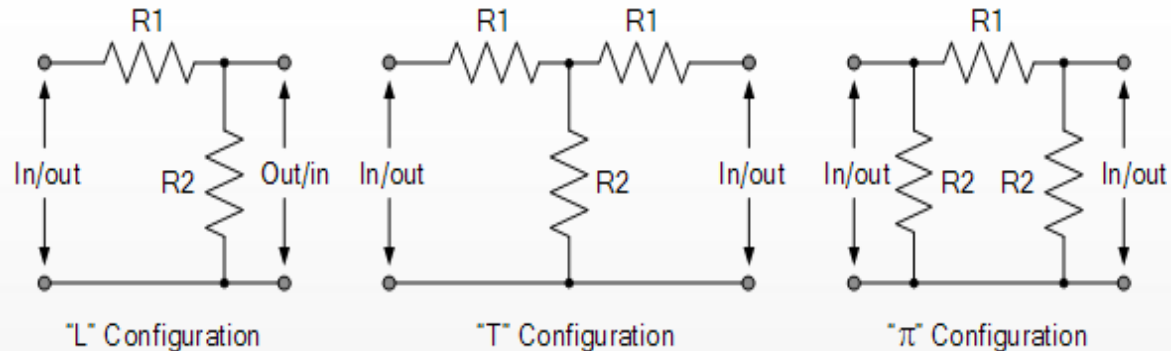
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1. Fixed Type Attenuator



- ✓ *In Fixed type attenuators the resistor network is locked at a predetermined attenuation value.*
- ✓ *These are laid down in the signal path to attenuate the power of the transmitted signal.*
- ✓ *These can be unidirectional or bidirectional based on their application requirement.*
- ✓ *This resistance value depends on the dimensions of the chip and the materials used for chip production.*

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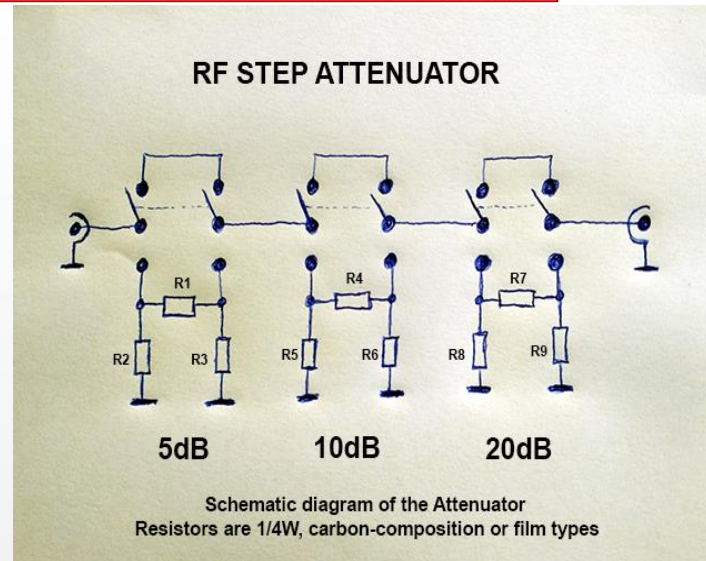
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2. Step Type Attenuator



- ✓ These attenuators are similar to fixed attenuators.
- ✓ But in this type, a push-button is provided to adjust the attenuation values. These provide attenuation values only from the pre-calibrated steps.

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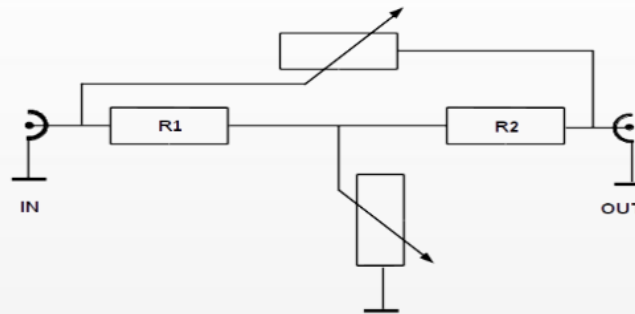
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3. Continuously Variable Type Attenuator



- ✓ *These attenuators are similar to fixed attenuators.*
- ✓ *But in this type, a push-button is provided to adjust the attenuation values. These provide attenuation values only from the pre-calibrated steps.*

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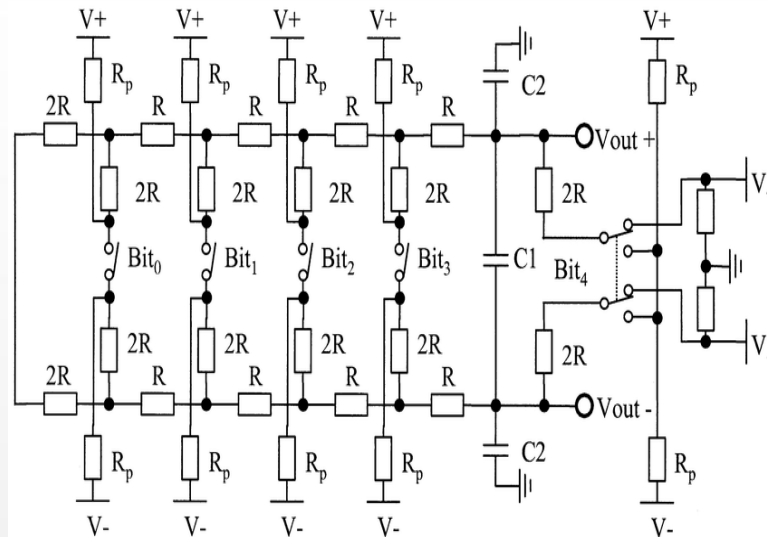
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4. Programmable Type Attenuator



- ✓ This type is also popularly called as the 'Digital step attenuator'.
- ✓ component is controlled by a computer-driven external control signal.

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5. Bias Type Attenuator

- ✓ *This type of attenuator has capacitance both at the input port and output port of the device which blocks the DC voltages.*
- ✓ *Thus, apart from attenuating the RF signals, this type passes the DC signals.*

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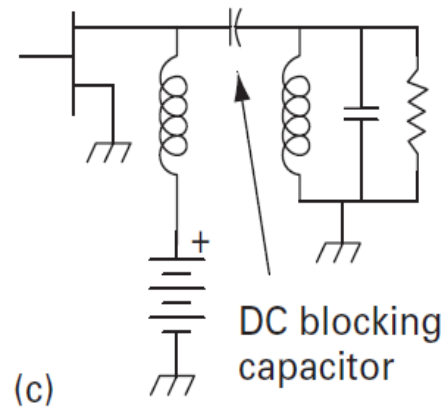
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6. DC Blocking Type Attenuator



✓ *This type is similar to the DC Bias type. The only difference between these two is the way the DC signal is totally blocked without any alternative path running towards the output port.*

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7. Optical Attenuators

- ✓ *These are similar to RF attenuator but instead of electrical signals, these attenuate the light waves.*
- ✓ *This attenuator absorbs or dissipates light according to the attenuation values without changing the waveform.*
- ✓ *Similar to the RF attenuators, optical attenuators are also designed as fixed, variable, programmable, etc....These are designed based on the application requirement.*

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- ✓ Basic circuits used in attenuators are pi pads (π -type) and T pads.
- ✓ These may be required to be balanced or unbalanced networks depending on whether the line geometry with which they are to be used is balanced or unbalanced.
- ✓ For instance, attenuators used with coaxial lines would be the unbalanced form while attenuators for use with twisted pair are required to be the balanced form.

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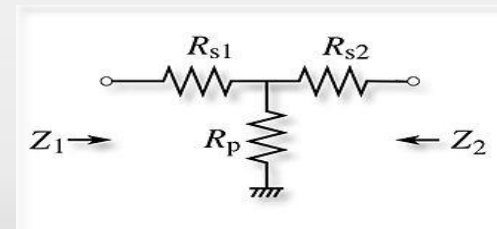
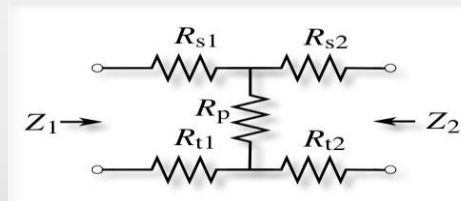
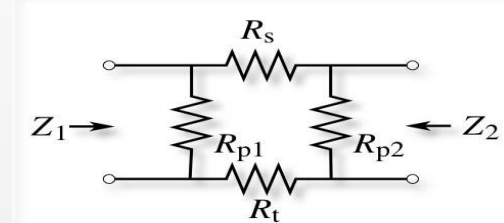
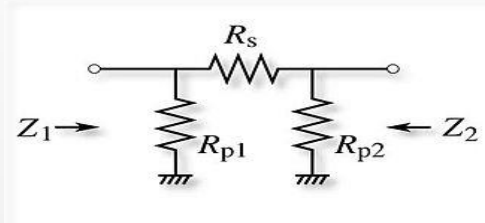
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- ✓ Four fundamental attenuator circuit diagrams are given: Pi type unbalanced attenuator circuit, Pi type balanced attenuator circuit, T type balanced attenuator circuit, T type unbalanced attenuator circuit respectively.



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- ✓ Attenuation expressed in decibels of relative power. A 3 dB pad reduces power to one half
- ✓ Nominal impedance, for example 50 ohm
- ✓ Frequency bandwidth, for example DC-18 GHz
- ✓ Accuracy
- ✓ Repeatability

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- ✓ *For testing purposes in laboratories, to obtain smaller voltage signals, attenuators are used.*
- ✓ *Fixed attenuators are used to improve the impedance matching in circuits.*
- ✓ *Optical attenuators are applied in fiber optic communication to properly match transmitter and receiver levels.*
- ✓ *To control volume of electronic equipment such as speakers, electric guitars, etc. attenuators are used.*
- ✓ *In some digital or analog circuits, a surge in voltage may cause damage to the circuits which can be avoided using attenuators to reduce high voltages.*

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Questions and Discussion

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End of Lecture

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Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

Good Luck
See You in
Next Lecture . . .



With My Best Wishes
Dr. Eng. Mohamed Abdo